Individual Reflection

Ray Chandonnet
7/28/2025

I intentionally waited until we were mostly done with the EDA before writing this - as of Friday at midnight which is when this was due, we were still DEEP into trying to pull our data directly into R by using the API's available from our three primary data sources. It's an important consideration for our study since we are using time series data, so any time we want to rerun the analysis with live data it's good to have that fully integrated. So I would say that certainly one of my biggest takeaways is that making the data integration dynamic was more challenging (yet rewarding) than we expected. Getting the API keys, connecting and pulling data via Python was straightforward enough (though my partner did the heavy lifting). What was not as straightforward is that we have opted to use R for our EDA because of the better tooling available through ggplot2 (more on that in a bit). We didn't want to have to collect the data via API in Python and then save the data in static csv files which seemed inelegant. So we had to learn how to run Python code within R using the "reticulate" package / library and functions. We were successful - the data is now sourced via API directly into R dataframes - but it used up quite a bit of time.

I guess the second set of takeaways related to the EDA exercise and results itself. Honestly, I learned so much more about the data that will meaningfully impact our project than I ever expected. I have not loved doing EDA in previous courses / projects and felt like it was a bit of "make work" - but it turns out I just wasn't doing it right in the past. In this situation I was surprised at how integral the results of well-designed EDA will be. For example - I had no idea how important the "missingness" analysis is to assess sparsity in my data set until I visualized it a few different ways and the results were a revelation. The correlation and distribution analyses were equally helpful to nail down how we might control / segment our data, to determine which features are most likely to be useful to our predictive models, to "frame" our hypothesis and set us up so we know right away what will need to be improved on to "prove" it, and to identify correlated feature variables that can be condensed to reduce dimensionality and collinearity's contribution to model variance. I must say - this is an area where Generative AI was really helpful - because I didn't have a strong foundation in EDA, I constructed a well-designed GenAI prompt describing our hypothesis and data set, and asked for recommendations to augment my own thoughts on what EDA to do. The results were incredibly helpful, as was ChatGPT's helpful hints on using ggplot - the EDA visualizations that came out of this exercise are absolute game changers.

One last thing - my partner and I continue to be very much in synch and so there have been no major disagreements including during the EDA phase.